HW2.1

SCIE 001 Math

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1. Suppose that f is differentiable with $f'(x) > 0$ for all x and suppose that $f(1) = 0$. Set

$$F(x) = \int_0^x f(t)\,\mathrm{d}t$$

Determine if each statement is True or False. Justify your answer.

(a) The function $F(x)$ is continuous.

(b) The function $F(x)$ is not twice differentiable.

(c) The value x = 1 is a critical point of F.

(d) The function $F(x)$ takes on a local minimum at $x = 1$.

(e) $F(1) > 0$.

Based on your answers above, make a rough sketch of the graph of $F(x)$.

2. Recall in class we proved that $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. In this problem, we will construct a proof of $S = \sum_{i=1}^{k} i^2$

(a) Calculate the value of S for $k = 1, 2, 3$ and 4.

$$S_{k=1} = \sum_{i=1}^{1} i^2 = 1^2 = 1$$

$$S_{k=2} = \sum_{i=1}^{2} i^2 = 1^2 + 2^2 = 3$$

$$S_{k=3} = \sum_{i=1}^{3} i^2 = 1^2 + 2^2 + 3^2 = 14$$

$$S_{k=4} = \sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30$$

(a) Calculate the value of $\frac{k(k+1)(2k+1)}{6}$

(b) Show that $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ by constructing a proof by induction by:

i. showing the equality is true for the base case (i.e., $k = 1$).

ii. for each positive integer N, the equality holds for $k = N$ implies the equality for $k = N + 1$.

Be sure to end your proof with an appropriate conclusion statement (i.e., "Therefore, by induction, ...").

3. Let $f(x)$ be an integrable function over the interval $[a, b]$. Choose an integer N and let

$$x_k = a + k\Delta x, \quad k = 0, 1, 2, ..., N \text{ where } \Delta x = \frac{b-a}{N}$$

The trapezoid rule is an approximation of the definite integral:

$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2},$$

where $\Delta x = \frac{b-a}{N}$

A bound on the error is given by the error formula

$$\left| \int_a^b f(x)\,\mathrm{d}x - \frac{b-a}{N} \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2} \right| \leq \frac{(b-a)^3}{12N^2} K_2$$

where K_2 is any number such that $|f''(x)| \leq K_2$ for all $x \in [a, b]$.

Consider the Fresnel integral

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\,\mathrm{d}x$$

(a) Write out and calculate the Left and Right Riemann sums for $N = 4$.

(b) Write our and calculate the Trapezoid rule for $N = 4$.

(c) Find and appropriate value of K_2 for the Fresnel integral. Be sure to justify your answer.

(d) Find the value of (the smallest) N which guarantees the trapezoid rule approximates the Fresnel Integral with errors less than 10^{-1} and 10^{-2}. Be sure to justify your answers.

4. In class, we proved Part 2 of the Fundamental Theorem of Calculus, which states that if f is a continuous function on the interval $[a, b]$ and F is any antiderivative of f, then

$$F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$$

Here, you will construct another proof of Part 2 of the Theorem.

(a) Divide the interval $[a, b]$ in n subintervals with endpoints $a = x_0 < x_1 < x_2 < ... < x_n = b$. Show that

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$

(b) Now suppose that F is any antiderivative of f. Show that there exists a number c_i in each interval $[x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$$

(c) Now build an appropriate Riemann sum for f on $[a, b]$ and show that

$$F(b) - F(a) = \int_a^b f(x) \, \mathrm{d}x$$

A Guide to Finding Intersection Points of P_{in} and P_{out} in Python

Create a new Jupyter notebook using anaconda. Copy and paste the following lines of code into a new cell.

1. **<u>Your task:</u>** Derive the EBM given above and verify that the units on both sides of the

Left Side	**Right Side**
$= JK^{-1} \frac{K}{s}$	$= W - W$
$= \frac{J}{s}$	$= W$
$= W$	

$$W = W$$

Left Side = Right Side

Therefore, the units are consistent.

$$\lim_{T \to 247^-} \alpha(T) = \alpha(247) \qquad \lim_{T \to 247^+} \alpha(T) = \alpha(247) \qquad \lim_{T \to 282^-} \alpha(T) = \alpha(282) \qquad \lim_{T \to 282^+} \alpha(T) = \alpha(282)$$